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BTech Cse, 3rd year

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Experiment No. 9

Experiment Name

Data Storytelling and Business Insight Generation using Interactive Visualizations

Aim

To create a compelling data-driven story by analyzing a real-world dataset, developing meaningful visualizations, and presenting actionable business insights to support strategic decision-making.

GITHUB LINK: https://github.com/Abhay34501/Mastery-in-Data-Analytics-and-Visualizations-and-Career-Advancement/blob/main/Experiment%209.py

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

sns.set_theme(style="whitegrid")

url = "https://raw.githubusercontent.com/yannie28/Global-Superstore/master/Global_Superstore(CSV).csv"

df = pd.read_csv(url, encoding="utf-8-sig")

print("Dataset Loaded Successfully!")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

df.head()
```

```
Dataset Loaded Successfully!
Rows: 1000
Columns: 24
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Postal Code	City	...	Product ID	Category	Sub-Category	Product Name
0	40098	CA-2014-AB10015140-41954	11/11/2014	11/13/2014	First Class	AB-100151402	Aaron Bergman	Consumer	73120.0	Oklahoma City	...	TEC-PH-5816	Technology	Phones	Samsung Convoy 3
1	26341	IN-2014-JR162107-41675	2/5/2014	2/7/2014	Second Class	JR-162107	Justin Ritter	Corporate	NaN	Wollongong	...	FUR-CH-5379	Furniture	Chairs	Novimex Executive Leather Armchair, Black
2	25330	IN-2014-CR127307-41929	10/17/2014	10/18/2014	First Class	CR-127307	Craig Reiter	Consumer	NaN	Brisbane	...	TEC-PH-5356	Technology	Phones	Nokia Smart Phone, with Caller ID
3	13524	ES-2014-KM1637548-41667	1/28/2014	1/30/2014	First Class	KM-1637548	Katherine Murray	Home Office	NaN	Berlin	...	TEC-PH-5267	Technology	Phones	Motorola Smart Phone, Cordless
4	47221	SG-2014-RH9495111-41948	11/5/2014	11/6/2014	Same Day	RH-9495111	Rick Hansen	Consumer	NaN	Dakar	...	TEC-CO-6011	Technology	Copiers	Sharp Wireless Fax, High-Speed

5 rows × 24 columns

```
print("Dataset Shape:", df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)
```

```
Dataset Shape: (1000, 24)

Column Names:
['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Postal Code', 'City', 'State', 'Country

Data Types:
Row ID              int64
Order ID           object
```

```
Order Date         object
Ship Date          object
Ship Mode          object
Customer ID        object
Customer Name      object
Segment            object
Postal Code       float64
City               object
State              object
Country            object
Region             object
Market             object
Product ID         object
Category           object
Sub-Category       object
Product Name       object
Sales             float64
Quantity            int64
Discount          float64
Profit            float64
Shipping Cost     float64
Order Priority     object
dtype: object
```

```
print("Missing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:", df.duplicated().sum())
```

```
Missing Values:
Row ID              0
Order ID            0
Order Date          0
Ship Date           0
Ship Mode           0
Customer ID         0
Customer Name       0
Segment             0
Postal Code       806
City                0
State               0
Country             0
Region              0
Market              0
Product ID          0
Category            0
Sub-Category        0
Product Name        0
Sales               0
Quantity            0
Discount            0
Profit              0
Shipping Cost       0
Order Priority      0
dtype: int64

Duplicate Rows: 0
```

```
df["Order Date"] = pd.to_datetime(df["Order Date"], errors="coerce")
df["Ship Date"] = pd.to_datetime(df["Ship Date"], errors="coerce")

for col in ["Sales", "Quantity", "Discount", "Profit", "Shipping Cost"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")

df = df.drop_duplicates()

df["Year"] = df["Order Date"].dt.year
df["Month"] = df["Order Date"].dt.month

print("Data Cleaning Completed!")
print("New Shape:", df.shape)
```

```
Data Cleaning Completed!
New Shape: (1000, 26)
```

```
df[[
    "Sales",
    "Quantity",
    "Discount",
    "Profit",
    "Shipping Cost"
]].describe().round(2)
```

	Sales	Quantity	Discount	Profit	Shipping Cost
count	1000.00	1000.00	1000.00	1000.00	1000.00
mean	1710.97	5.56	0.09	288.92	272.38
std	1259.24	2.72	0.15	574.50	176.16
min	1.91	1.00	0.00	-3059.82	1.07
25%	826.91	4.00	0.00	10.04	209.83
50%	1585.12	5.00	0.00	190.68	258.90
75%	2477.81	7.00	0.15	518.87	351.07
max	9892.74	14.00	0.80	4946.37	923.63

```
total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
total_orders = df["Order ID"].nunique()
total_customers = df["Customer ID"].nunique()

profit_margin = (total_profit / total_sales) * 100
average_order_value = total_sales / total_orders

print("========== KPIs ==========")
print("Total Sales       :", round(total_sales, 2))
print("Total Profit      :", round(total_profit, 2))
print("Total Orders      :", total_orders)
print("Total Customers   :", total_customers)
print("Average Order Value:", round(average_order_value, 2))
print("Profit Margin     :", round(profit_margin, 2), "%")
```

```
========== KPIs ==========
Total Sales       : 1710971.47
Total Profit      : 288920.44
Total Orders      : 867
Total Customers   : 800
Average Order Value: 1973.44
Profit Margin     : 16.89 %
```

```
yearly = df.groupby("Year")[["Sales", "Profit"]].sum().reset_index()

display(yearly.round(2))

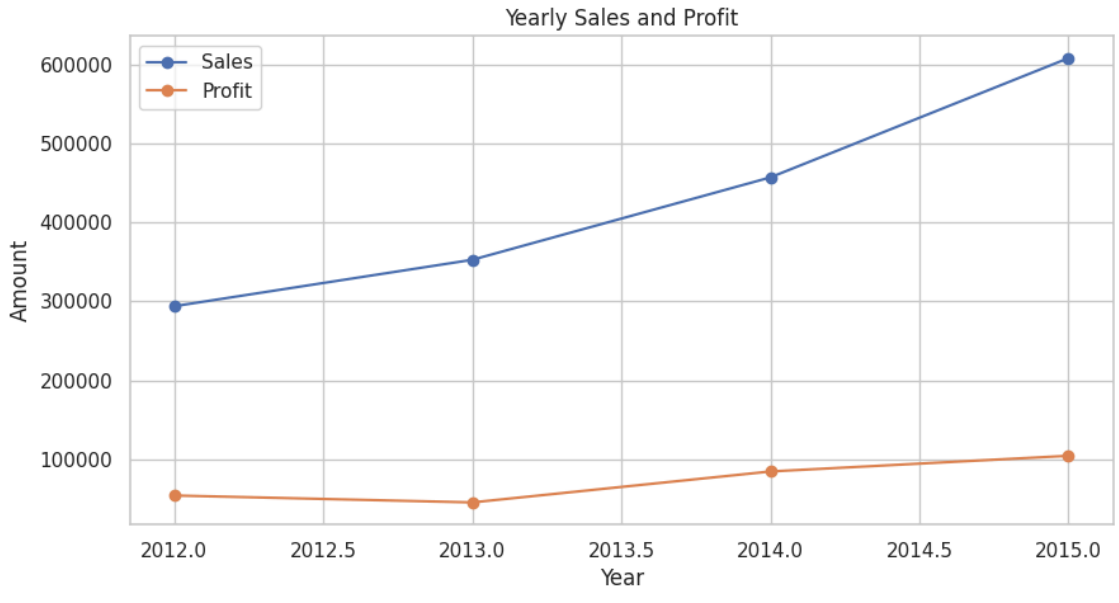
plt.figure(figsize=(10,5))

plt.plot(yearly["Year"], yearly["Sales"],
         marker="o", label="Sales")

plt.plot(yearly["Year"], yearly["Profit"],
         marker="o", label="Profit")

plt.title("Yearly Sales and Profit")
plt.xlabel("Year")
plt.ylabel("Amount")
plt.legend()
plt.show()
```

	Year	Sales	Profit
0	2012	293829.82	54249.43
1	2013	352570.33	45515.51
2	2014	456844.29	84663.14
3	2015	607727.03	104492.36



```
region = df.groupby("Region")[["Sales", "Profit"]].sum().reset_index()

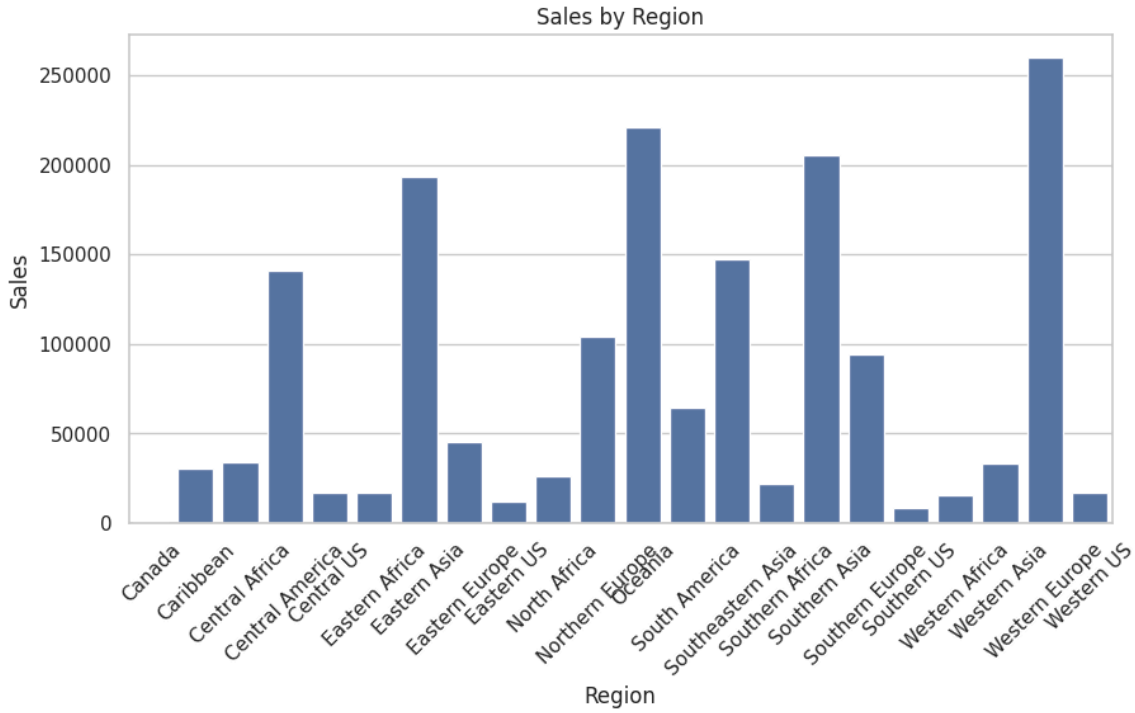
display(region.round(2))

plt.figure(figsize=(10,5))

sns.barplot(data=region, x="Region", y="Sales")

plt.title("Sales by Region")
plt.xticks(rotation=45)
plt.show()
```

	Region	Sales	Profit
0	Canada	887.01	213.72
1	Caribbean	30746.74	5463.82
2	Central Africa	33810.48	9668.22
3	Central America	140966.21	21741.95
4	Central US	17187.94	5676.02
5	Eastern Africa	17387.94	4302.72
6	Eastern Asia	193590.60	37911.16
7	Eastern Europe	45662.19	12228.45
8	Eastern US	12256.44	2544.89
9	North Africa	26266.44	7416.42
10	Northern Europe	103888.63	23552.73
11	Oceania	220809.08	33777.56
12	South America	64550.54	7723.58
13	Southeastern Asia	147337.60	10561.80
14	Southern Africa	22262.28	4876.32
15	Southern Asia	205466.26	37536.60
16	Southern Europe	94441.11	20557.50
17	Southern US	8360.75	1384.72
18	Western Africa	15478.79	1278.17
19	Western Asia	33308.21	5262.35
20	Western Europe	259576.28	32783.40
21	Western US	16729.95	2458.34



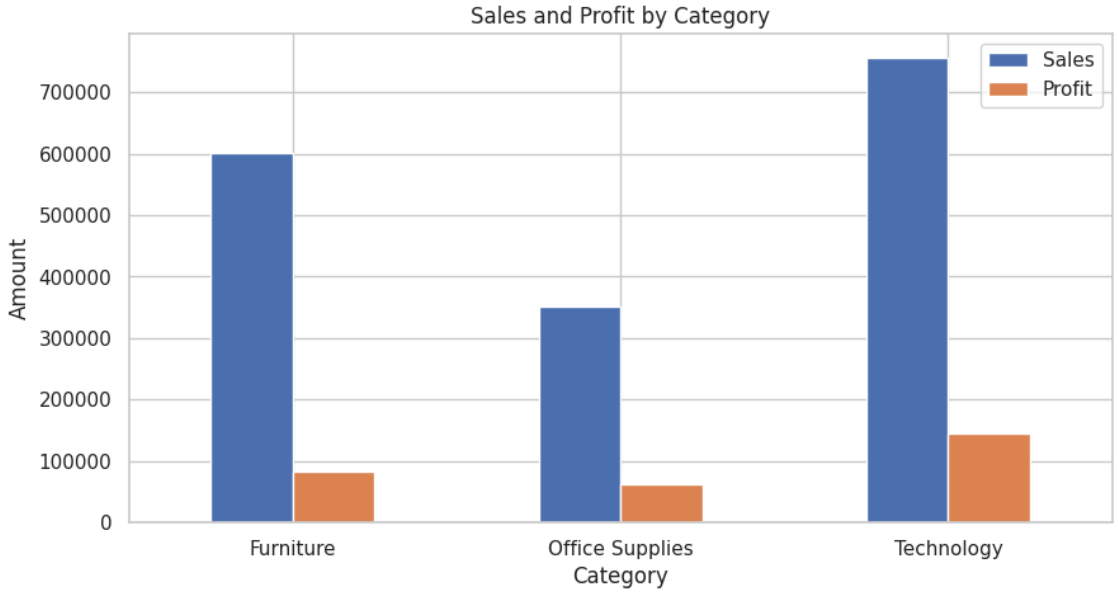

```
category = df.groupby("Category")[["Sales", "Profit"]].sum().reset_index()

display(category.round(2))

category.plot(
    x="Category",
    y=["Sales", "Profit"],
    kind="bar",
    figsize=(10,5)
)
```

```
plt.title("Sales and Profit by Category")
plt.ylabel("Amount")
plt.xticks(rotation=0)
plt.show()
```

	Category	Sales	Profit
0	Furniture	601492.31	82188.95
1	Office Supplies	352496.14	61662.81
2	Technology	756983.02	145068.68




```
best_region = region.loc[region["Sales"].idxmax()]
best_category = category.loc[category["Sales"].idxmax()]
profit_category = category.loc[category["Profit"].idxmax()]
worst_region = region.loc[region["Profit"].idxmin()]

print("========== KEY INSIGHTS ==========\n")

print("1. Highest Sales Region:",
      best_region["Region"])

print("2. Highest Sales Category:",
      best_category["Category"])

print("3. Highest Profit Category:",
      profit_category["Category"])

print("4. Lowest Profit Region:",
      worst_region["Region"])

print("\nOverall Profit Margin:",
      round(profit_margin, 2), "%")
```

```
========== KEY INSIGHTS ==========

1. Highest Sales Region: Western Europe
2. Highest Sales Category: Technology
3. Highest Profit Category: Technology
4. Lowest Profit Region: Canada

Overall Profit Margin: 16.89 %
```

Experiment 9 — Question Answers

Q1. What is data storytelling? How is it different from data visualization?

Answer: Data storytelling is the process of combining **data, visualization, and narrative** to communicate meaningful insights effectively. Data visualization mainly represents data using charts and graphs, whereas data storytelling explains the meaning and importance of those visualizations and connects them with business decisions.

Q2. Why is storytelling important in business analytics and decision-making?

Answer: Storytelling makes complex analytical results easier to understand. It helps stakeholders identify important trends, problems, opportunities, and risks. It also connects data findings with business objectives and supports better decision-making.

Q3. What are the essential components of an effective data story?

Answer: The essential components are:

1. Clear business problem or objective

2. Relevant and reliable data
3. Data analysis
4. Meaningful visualizations
5. Key findings and insights
6. Context and explanation

Q4. How do Key Performance Indicators (KPIs) enhance business reporting?

Answer: KPIs provide a quick summary of important business performance measures. Examples include **Sales, Profit, Number of Orders, Average Order Value, and Profit Margin**. They help managers monitor performance, compare results, identify gaps, and make informed decisions.

Q5. Why should visualizations be arranged in a logical sequence while presenting analytical findings?

Answer: Visualizations should be arranged logically so that the audience can easily follow the data story. A proper sequence can move from **problem → data → analysis → findings → recommendations**. This makes the presentation clear, organized, and easier to understand.

Q6. What factors should be considered while selecting visualizations for a business presentation?

Answer: The following factors should be considered:

- Type of data
- Purpose of analysis
- Comparison required
- Number of categories
- Audience
- Readability

Q7. Explain how dashboards and storytelling complement each other in Business Intelligence.

Answer: Dashboards provide an interactive view of important KPIs, trends, and business metrics. Storytelling explains the meaning of these results and highlights the most important findings. Therefore, dashboards provide the **visual evidence**, while storytelling provides the **context and explanation**.

Q8. What challenges may arise while communicating analytical insights to non-technical stakeholders?

Answer: Some common challenges are:

1. Complex technical terminology
2. Too many charts or excessive information
3. Lack of business context
4. Difficulty understanding statistical concepts

Q9. Give two real-world examples where data storytelling has influenced business or policy decisions.

Answer:

Example 1 — Retail: Retail companies use sales dashboards and data stories to identify product demand, sales trends, and regional performance. This information can support decisions about inventory, promotions, and product planning.

Q10. How can effective data storytelling improve strategic planning and organizational performance?

Answer: Effective data storytelling helps organizations understand their current performance, identify trends and performance gaps, and communicate findings clearly to stakeholders. It supports better **planning, resource allocation, monitoring, and decision-making**, which can contribute to improved organizational performance.
